

Python & PyQt5 for the Python version of the Beboputer Computer Simulator

Installation Guide

Windows • Linux • macOS

Introduction

This guide walks you through installing Python 3 and PyQt5 on Windows, Linux, and macOS. PyQt5 is a set of Python bindings for Qt5, desktop GUI applications.

Prerequisites: a working internet connection and basic familiarity with your operating system's terminal or command prompt.

Quick Reference

Use the table below to find the key commands for your platform at a glance.

Platform	Verify Python	Install PyQt5	Test Import
Windows	<code>python --version</code>	<code>pip install PyQt5</code>	<code>python -c "import PyQt5"</code>
Linux	<code>python3 --version</code>	<code>pip3 install PyQt5</code>	<code>python -c "import PyQt5"</code>
macOS	<code>python3 --version</code>	<code>pip3 install PyQt5</code>	<code>python -c "import PyQt5"</code>

Windows

Windows Installation

Step 1 — Install Python

1. Open your browser and go to <https://www.python.org/downloads/>
2. Click the Download Python 3.x.x button (latest stable release).
3. Run the downloaded installer.
4. Important: on the first screen, tick Add Python to PATH before clicking Install Now.
5. Click Install Now and wait for the installation to complete.

 **Important:** Always check “Add Python to PATH” during installation. Without this, Python commands will not work in Command Prompt.

Step 2 — Verify Python

Open Command Prompt (press Win+R, type cmd, press Enter) and run:

```
python --version
python -m pip --version
```

You should see output similar to: Python 3.12.x and pip 24.x.x

Step 3 — Install PyQt5

In Command Prompt, run:

```
pip install PyQt5
```

pip will download and install PyQt5 and its dependencies automatically.

Step 4 — Verify PyQt5

```
python -c "import PyQt5; print('PyQt5 OK')"  
python -c "from PyQt5.QtWidgets import QApplication; print('Widgets OK')"
```

✓ **Success:** If you see PyQt5 OK and Widgets OK, installation is complete.

Linux



Linux Installation

Instructions below cover Debian/Ubuntu-based distributions (apt), Fedora/RHEL (dnf), and Arch Linux (pacman).

Step 1 — Install Python

Debian / Ubuntu / Linux Mint

```
sudo apt update  
sudo apt install python3 python3-pip python3-venv -y
```

Fedora / RHEL / CentOS Stream

```
sudo dnf install python3 python3-pip -y
```

Arch Linux / Manjaro

```
sudo pacman -Sy python python-pip
```

Step 2 — Verify Python

```
python3 --version  
pip3 --version
```

Step 3 — Install PyQt5

Option A — Via pip (recommended for all distros)

```
pip3 install PyQt5
```

Option B — Via system package manager

Debian / Ubuntu:

```
sudo apt install python3-pyqt5 -y
```

Fedora:

```
sudo dnf install python3-qt5 -y
```

Arch Linux:

```
sudo pacman -Sy python-pyqt5
```

 **Tip:** Using a virtual environment (`python3 -m venv .venv`) is recommended to avoid conflicts with system packages.

Step 4 — Verify PyQt5

```
python3 -c "import PyQt5; print('PyQt5 OK')"  
python3 -c "from PyQt5.QtWidgets import QApplication; print('Widgets OK')"
```

Display Server Note

PyQt5 requires a display server. On headless servers (no GUI), you may need to install a virtual framebuffer:

```
sudo apt install xvfb -y  
Xvfb :99 -screen 0 1024x768x24 &  
export DISPLAY=:99
```

macOS



macOS Installation

Step 1 — Install Homebrew (recommended)

Homebrew is the most convenient way to manage Python on macOS. Open Terminal (Cmd+Space, type Terminal) and run:

```
/bin/bash -c "$(curl -fsSL
https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```



Tip: If you already have Python installed via the python.org installer, you can skip Homebrew and use that instead.

Step 2 — Install Python

Via Homebrew (recommended)

```
brew install python
```

Via python.org installer

Download and run the macOS installer from <https://www.python.org/downloads/> — it will add Python to your PATH automatically.

Step 3 — Verify Python

```
python3 --version
pip3 --version
```

Step 4 — Install PyQt5

```
pip3 install PyQt5
```

Alternatively, using Homebrew:

```
brew install pyqt@5
```

⚠ **Apple Silicon (M1/M2/M3):** PyQt5 via pip works on Apple Silicon. If you encounter issues, try installing under Rosetta or use a conda environment with `conda install pyqt`.

Step 5 — Verify PyQt5

```
python3 -c "import PyQt5; print('PyQt5 OK')"  
python3 -c "from PyQt5.QtWidgets import QApplication; print('Widgets OK')"
```

Using Virtual Environments (Recommended)

Virtual environments isolate your project's dependencies from the system Python. This prevents version conflicts and is considered best practice.

Create and Activate

Windows

```
python -m venv .venv  
.venv\Scripts\activate  
pip install PyQt5
```

Linux / macOS

```
python3 -m venv .venv  
source .venv/bin/activate  
pip install PyQt5
```

Once activated, your prompt will be prefixed with `(.venv)`. Use `deactivate` to exit the environment.

Hello World — Testing Your Installation

Create a file called `hello.py` with the following content and run it to confirm everything is working:

```
import sys
from PyQt5.QtWidgets import QApplication, QLabel, QWidget

app = QApplication(sys.argv)

window = QWidget()
window.setWindowTitle('Hello PyQt5')
window.resize(300, 100)

label = QLabel('PyQt5 is working!', window)
label.move(80, 35)

window.show()
sys.exit(app.exec_())
```

Run it

Windows

```
python hello.py
```

Linux / macOS

```
python3 hello.py
```

A small window titled “Hello PyQt5” should appear. If it does, your environment is fully set up.

To run the Beboputer

```
python3 Beboputer.py
```

Troubleshooting

'python' is not recognised (Windows)

Python was not added to PATH during installation. Fix options:

- Re-run the installer and tick Add Python to PATH.
- Or add Python manually: search for Environment Variables in Windows settings, then add the Python install path to the Path variable.

pip: command not found (Linux / macOS)

```
# Try pip3 instead:
pip3 install PyQt5

# Or use Python module syntax:
python3 -m pip install PyQt5
```

Cannot import PyQt5

This usually means PyQt5 was installed into a different Python interpreter. Check which Python pip is targeting:

```
# Windows
where python
where pip

# Linux / macOS
which python3
which pip3
```

If they point to different locations, use:

```
python3 -m pip install PyQt5
```


macOS: xcrun error or missing Command Line Tools

```
xcode-select --install
```

Permission denied errors (Linux)

Avoid sudo pip. Use a virtual environment or the --user flag instead:

```
pip3 install --user PyQt5
```

 **Useful links:** Python downloads: <https://www.python.org/downloads> | PyQt5 docs: <https://www.riverbankcomputing.com/static/Docs/PyQt5> | Qt5 docs: <https://doc.qt.io/qt-5>